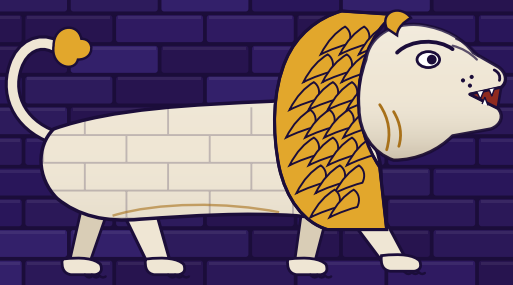
PHYSICS & ENERGY TECHNOLOGY · LEVEL 3

PD

PROFESSIONAL DEVELOPMENT SEMINAR

A handout of ideas, facts and problems



8 sections · Edition of November 2026 · 26 pages

Contents

How to read this handout . . . 1

Professional Development Seminar . . . 2

1 Teaching as a profession . . . 2
2 Professionalization and the teacher role . . . 3
3 Teacher formation: before and during service . . . 4
4 Preparation, practice and professional standards . . . 6
5 Plan continuing professional development . . . 7
6 Portfolios and reflective evaluation . . . 8
7 Ethics, collaboration and learning communities . . . 10
8 Educational technology and evaluation of impact . . . 11

Hints . . . 13

Answers . . . 14

Professional Development Seminar, Test A . . . 16

Professional Development Seminar, Test B . . . 17

Professional Development Seminar, Test A: answers . . . 18

Professional Development Seminar, Test B: answers . . . 19

Facts at a glance . . . 20

Notation . . . 21

Sources . . . 22

Index . . . 23

Index of notation . . . 24

How to read this handout

This handout follows the method of Jaan Kalda's physics handouts. The text is short; the work is in the numbered items. Read each section once, then return to its items until you can use them without looking.

Idea 1. An *idea* is a way of thinking that unlocks a whole family of problems. Ideas are framed in violet.

Fact 1. A *fact* is a law or relationship you are expected to know, with the meaning of every symbol and the conditions under which it holds.

Method 1. A *method* is a fixed sequence of steps for a standard task.

Pr 1. Problems are numbered through the handout. A problem followed by a *Solution* is worked for you; cover the solution and try first. For the remaining problems, attempt the problem, then use the *Hints* section, and only then the *Answers* section, both at the back of the subject.

Physical quantities are set in italic, units in roman, with a thin space between number and unit. The index at the end lists every topic and symbol with the page on which it is introduced.

Professional Development Seminar

Elective lecture themes: teaching as a profession, teacher formation, professional learning, portfolios, standards and educational technology. This follows the teacher-development focus of the supplied file; it is not a generic career-skills course.

1 Teaching as a profession

Aims. Explain two criteria that make teaching a profession; Support each criterion with a distinct classroom example.
Before this section: A professional criterion is a defining feature, not simply an employment benefit.

The source distinguishes a profession through specialized knowledge, extended preparation, social purpose, ethical responsibilities and organized practice. Teaching is more than delivering facts: it develops learners and contributes to society. A strong exam response connects each professional criterion to a teaching example.

Idea 1. A profession combines expertise with responsibility for how that expertise affects others. Teaching therefore involves both knowing a subject and making informed, ethical decisions about learners. An exam answer becomes convincing when it connects each professional feature to an observable teaching action.

Fact 1.

$$\text{Criterion} + \text{explanation} + \text{teaching example}$$

Why it holds. Specialized preparation explains why professional judgement requires more than familiarity with subject facts. Ethical responsibilities explain why competence must be used fairly and for a social purpose.

Method 1.

1. Define profession using expertise, preparation, social purpose and ethical responsibility.
2. Select two distinct criteria relevant to the question.
3. Explain each criterion in plain language.
4. Attach a concrete teaching example to each explanation.

Recognising it. For reasons teaching is a profession, give distinct criterion-example pairs instead of repeated general praise.

Pr 1. Specialized knowledge includes subject expertise and pedagogy; ethical responsibility includes fair assessment and respect for learners.

Solution. Subject expertise concerns what is taught; pedagogy concerns how learners are helped to understand it. Together they illustrate specialized professional knowledge rather than information delivery alone. Fair assessment and respect for learners illustrate

ethical responsibility. The examples support two different criteria, so they provide more evidence than repeating expertise in several forms.

Common error. Do not assume being paid alone makes an activity a profession.

In the examination. Use two complete criterion-explanation-example pairs when four marks are available, rather than a long unsupported list.

Pr 2. Does being paid alone establish that an activity is a profession?

Pr 3. Give two reasons teaching is considered a profession, each with an example.

Pr 4. Which pair best shows two different professional criteria?

(a) Subject-and-pedagogy preparation, plus fair assessment

(b) Subject knowledge, plus knowing many subject facts

(c) Being busy, plus having a long working day

2 Professionalization and the teacher role

Aims. Describe teacher roles through observable professional actions; Explain how assessment evidence guides subsequent instruction.
Before this section: A role describes a function; an action shows how that function appears in practice.

The lecture describes development from pre-professional through semi-professional toward growing professionalization. Teacher roles extend from presenting knowledge to guiding learning, assessing progress, modelling values and supporting society. Historical labels describe development, not a reason to dismiss present expertise.

Idea 2. Teacher roles describe purposes expressed through actions. An assessor interprets evidence, a guide helps learners choose strategies, and a model demonstrates professional values. A role label alone leaves the reader guessing; an action and its effect show that the role is understood.

Fact 2.

$$\text{Role} \rightarrow \text{observable professional action}$$

Why it holds. The same classroom activity can serve different roles depending on its purpose and follow-up. Describing a learner effect connects a teacher action to its professional function.

Method 2.

1. Identify the role asked about, such as assessor or learning guide.
2. Describe a specific action the teacher takes in that role.
3. Explain how the action supports learners.

4. For historical comparisons, discuss changes in preparation or organization rather than dismissing current expertise.

Recognising it. When asked for roles beyond presenting content, identify an action that changes how learning is supported.

Pr 5. A teacher acting as an assessor uses evidence from student work to adapt instruction.

Solution. As an assessor, the teacher examines student work to identify what learners understand. If several answers show the same error, that evidence indicates a learning gap. Adapting instruction in response connects the assessor role to better teaching decisions. As a guide, the teacher supports reasoning through questions or strategies instead of supplying every answer.

Common error. A list of role names without explanation earns less evidence of understanding.

In the examination. After every role name add a verb and object: diagnoses a misconception, gives targeted feedback or guides strategy choice.

Pr 6. Why is gives a test alone a weak explanation of the assessor role?

Pr 7. Describe two teacher roles beyond explaining subject content.

Pr 8. A teacher asks a learner to compare two solution strategies rather than immediately giving an answer. This most directly shows:

(a) Avoiding all responsibility for learning

(b) Guiding reasoning

(c) Merely storing information

3 Teacher formation: before and during service

Aims. Distinguish pre-service preparation, induction and continuing development; Match a support activity to its career stage and purpose.
Before this section: Teachers develop before entering practice and throughout their careers.

Teacher formation is a continuing process. Pre-service preparation establishes knowledge and skills; induction supports entry to actual teaching; continuing professional development addresses changing needs and improves practice. Training is one method of development, not the whole process.

Idea 3. Teacher formation is a continuing sequence, but support has different purposes at different stages. Initial preparation builds foundations before entry; induction helps a new teacher use them in a real workplace. Comparing both timing and purpose makes the distinction clearer than treating one as simply more training.

Fact 3.

Pre-service → induction → continuing development

Why it holds. Pre-service learning builds foundations before entry to the profession. Induction bridges those foundations and the demands of early independent practice.

Method 3.

1. Locate the person before employment, in early practice or in established service.
2. Name the matching formation stage.
3. Explain the specific knowledge or support needed at that stage.
4. Describe how the activity supports practice rather than treating attendance as proof of improvement.

Recognising it. Stage-comparison questions require a timing statement and a purpose statement for each stage.

Pr 9. A mentor supporting a newly appointed teacher belongs to induction.

Solution. The example teacher is newly appointed and is entering real classroom practice. Mentor support at this point is induction, bridging initial preparation and workplace demands. Later peer observation and inquiry are examples of continuing development. The method may look similar across stages, but the career context and purpose determine the classification.

Common error. Graduation marks an entry point, not the end of professional learning.

In the examination. In a comparison give both timing and purpose for each stage; stating before and after alone is incomplete.

Pr 10. Is every teacher workshop automatically pre-service preparation?

Pr 11. Compare pre-service preparation with induction in two clear points.

Pr 12. A newly appointed teacher receives workplace mentoring while beginning classroom duties. Why is this induction rather than pre-service?

(a) Mentoring can never be used before employment

(b) All learning stops after induction

(c) It supports entry into actual professional practice

4 Preparation, practice and professional standards

Aims. Explain how supervised field experience links theory with practice; Construct a feedback cycle using observable evidence.
Before this section: Preparation includes subject knowledge, pedagogy and supervised teaching; standards describe expected practice.

Teacher preparation combines subject knowledge, pedagogical understanding and supervised practice. Field experience connects theory with real learners and classroom decisions. Professional standards make expectations explicit and support self-evaluation. Licensing is a recognition process whose current requirements must be checked with the responsible body.

Idea 4. Reflection becomes professional improvement when it leads to a tested change. An observed lesson supplies evidence, but evidence must be interpreted, used to revise practice and checked again. The next lesson is therefore part of the cycle, rather than a separate event after feedback ends.

Fact 4.

$$\text{Knowledge} \rightarrow \text{supervised application} \rightarrow \text{feedback} \rightarrow \text{improvement}$$

Why it holds. Specific evidence makes the improvement target clearer than a vague judgement. Checking the revised lesson reveals whether the chosen change actually helped.

Method 4.

1. Identify the teaching expectation or intended learner outcome.
2. Collect relevant evidence from observation and student work.
3. Choose one specific change suggested by that evidence.
4. Apply the change and evaluate the next attempt against the same expectation.

Recognising it. An improvement-cycle question needs evidence, a focused need, a targeted action and a follow-up evaluation.

Pr 13. After teaching a supervised lesson, a trainee reviews observations and student work, then revises the next lesson.

Solution. The trainee first teaches a supervised lesson, putting planned ideas into practice. Observation notes and student work show what occurred and what learners achieved. The trainee compares that evidence with the intended learning and revises the next lesson. The cycle is completed by checking whether the revision improves the next observed outcome.

Common error. Do not invent current licence requirements from an older lecture.

In the examination. When standards are mentioned, link one expectation to evidence; do not invent current licensing rules from a lecture.

Pr 14. Why does feedback without a subsequent change and evaluation leave the cycle incomplete?

Pr 15. Give a four-step improvement cycle for a trainee after an observed lesson.

Pr 16. After an observed lesson, a trainee changes the task but collects no follow-up evidence. What is missing?

(a) Evaluation of whether the change helped

(b) Any possibility of changing practice

(c) A requirement to repeat the original lesson unchanged

5 Plan continuing professional development

Aims. Write a professional-development plan linking need, goal, activity and evidence; Distinguish participation evidence from evidence of impact.
Before this section: A development goal should address an identified teaching or learning need.

Professional development is purposeful and sustained. It begins with a learning need, sets a clear goal, chooses suitable activities and evaluates effects on practice and learners. Workshops, peer observation, professional reading and collaborative inquiry can all contribute.

Idea 5. A development plan is a chain from a real teaching need to evidence of improvement. A workshop can supply ideas, but the plan also needs classroom application and a way to judge results. A focused goal makes it possible to tell whether the activity was useful.

Fact 5.

$$\text{Need} \rightarrow \text{goal} \rightarrow \text{activity} \rightarrow \text{evidence of impact}$$

Why it holds. An activity becomes relevant when it addresses the diagnosed need. Before-and-after learner evidence can show impact more directly than participation records alone.

Method 5.

1. Use classroom evidence to name one specific development need.
2. Write a focused goal and a realistic review period.
3. Choose a learning activity and classroom application that address the goal.
4. Compare relevant evidence before and after the application to judge progress.

Recognising it. When asked for a development plan, link need, goal, activity and evidence within a defined review period.

Pr 17. A teacher notices weak student explanations, studies questioning strategies, tries them for four weeks and compares the quality of student responses.

Solution. The teacher notices weak student explanations, identifying a specific need. Studying questioning strategies is the development activity, but study alone does not show impact. Trying the strategies for four weeks puts the learning into classroom practice. Comparing response samples before and after can reveal whether explanations became clearer and better supported.

Common error. Attendance certificates show participation, not necessarily improved teaching.

In the examination. Keep all four plan parts linked to the same need; an unrelated workshop does not satisfy a coherent development plan.

Pr 18. What does a workshop attendance certificate demonstrate directly?

Pr 19. Write a short development plan for a teacher whose students struggle to explain answers.

Pr 20. A teacher attends a questioning workshop. Which next step best tests its value for weak student explanations?

(a) Collect another certificate without classroom application

(b) Apply selected strategies and compare explanation samples over time

(c) Count the slides shown in the workshop

6 Portfolios and reflective evaluation

Aims. Select portfolio evidence for a defined goal or standard; Explain how reflection turns an artifact into evidence of development.

Before this section: An artifact is a document or sample; evidence requires an explanation of what it demonstrates.

A professional portfolio is a purposeful selection of evidence linked to goals and standards, with reflection on what the evidence demonstrates. It can include plans, student work, feedback and records of improvement. A large unorganized collection is less useful than a selective, explained one.

Idea 6. A portfolio is an argument supported by selected artifacts. A lesson plan can show intended alignment, while student work can show something about the outcome. Commentary connects the evidence to a professional claim and acknowledges what one artifact alone cannot prove.

Fact 6.

$$\text{Evidence} + \text{reflection} + \text{link to a goal}$$

Why it holds. An artifact becomes useful evidence when its relevance to a goal or standard is explained. Reflection connects intention, outcome and a next learning step rather than simply decorating a collection.

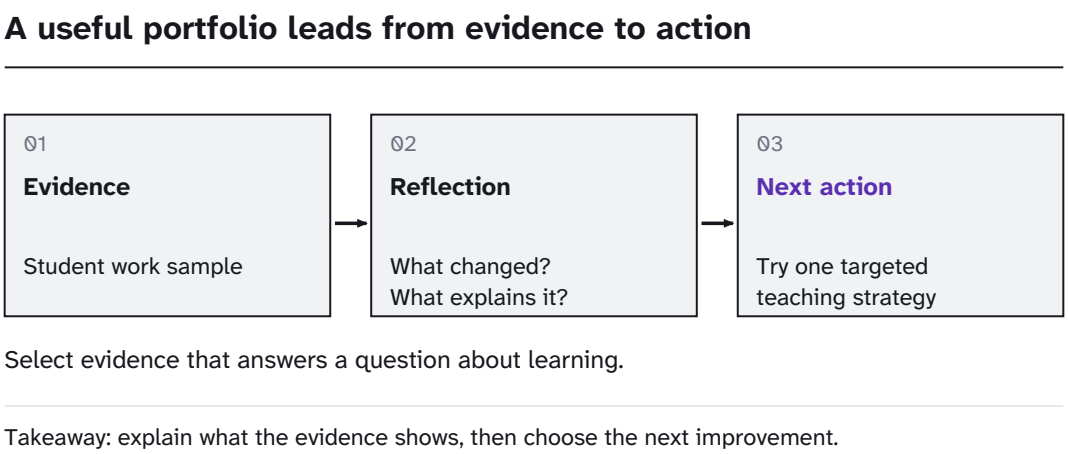


Figure 1. Figure 08. A portfolio connects selected evidence to reflection and a targeted next action. A student-work sample becomes more useful when the teacher explains what it shows and how teaching will change.

Method 6.

1. State the portfolio purpose or professional goal.
2. Select a small number of relevant artifacts and organize them clearly.
3. Explain what each artifact shows and what its limitations are.
4. Reflect on the outcome and identify a next improvement, protecting learner privacy.

Recognising it. When naming portfolio items, pair each with the specific aspect of practice it can demonstrate.

Pr 21. A lesson plan paired with student work and a reflection on changes shows more than a lesson plan alone.

Solution. The lesson plan shows what the teacher intended learners to do. Student work shows something about the resulting learning and where difficulties remained. A reflection compares the plan with that outcome and explains why a change was made. The three pieces together support a development claim more clearly than an unexplained plan alone.

Common error. Do not include identifiable student information unnecessarily; follow the institution privacy rules.

In the examination. For each item finish the sentence this demonstrates... and connect it to a goal or standard; anonymize learner material as required.

Pr 22. Does adding more unrelated documents necessarily make a portfolio stronger?

Pr 23. Name two portfolio items and explain what each can demonstrate.

Pr 24. A lesson plan alone most directly shows:

(a) That every student achieved the objective

(b) That no future improvement is needed

(c) What learning and teaching were intended

7 Ethics, collaboration and learning communities

Aims. Rewrite personal criticism as evidence-based constructive feedback; Connect ethical collaboration with learner respect and improvement.

Before this section: Observation is what can be seen or recorded; interpretation and personal judgement are different.

Professional learning can be collaborative: colleagues discuss evidence, share practice and solve problems. Ethical practice includes respect, fairness, honesty and responsibility. Feedback should focus on observable practice and improvement, rather than personal attacks.

Idea 7. Constructive feedback describes a changeable practice rather than labelling a person. The recipient needs to know what was observed, how learners were affected and what could be tried next. Respect and specificity make challenge more useful without requiring uncritical agreement.

Fact 7.

$$\text{Observation} + \text{effect} + \text{constructive next step}$$

Why it holds. Observable evidence gives the colleague something concrete to examine. An actionable suggestion turns criticism into a possible improvement step.

Method 7.

1. Identify an observable feature of teaching rather than a personal label.
2. Describe its apparent effect on learners with available evidence.
3. Suggest one feasible next step connected to that effect.
4. Use respectful language and follow confidentiality and reporting responsibilities.

Recognising it. To rewrite harsh feedback, replace personal judgement with observation, learner effect and a feasible next action.

Pr 25. A colleague gives feedback that a task confused learners and suggests clearer success criteria, rather than calling the teacher incompetent.

Solution. Calling a teacher incompetent attacks the person and gives no specific route to improvement. The example instead identifies that the task confused learners. Suggesting clearer success criteria connects the feedback to a teaching action the colleague can try. A

later check of learner understanding can show whether that change helped, completing the improvement conversation.

Common error. Collaboration is not simply agreeing with everyone or sharing private student details.

In the examination. Structure scenario answers as observation, learner effect and next step; avoid unsupported claims about personal character.

Pr 26. Is constructive feedback the same as agreeing with every colleague decision?

Pr 27. Rewrite this feedback constructively: Your lesson was bad.

Pr 28. Which statement is most useful as the observation part of feedback?

(a) Several learners asked what the task goal was

(b) You are a careless person

(c) Your lesson was bad in every way

8 Educational technology and evaluation of impact

Aims. Choose an educational tool to serve a specific learning objective; Evaluate technology through diagnosis, teacher response and learner evidence.

Before this section: A tool supports an activity; learning depends on how the activity and follow-up are designed.

Educational technology should serve a learning goal. Teachers need to select suitable tools, design learning activities, assess outcomes and consider access and ethical use. Technical operation alone does not establish teaching value. The seminar mentions digital competencies; programming instruction is excluded from this pack as requested.

Idea 8. Technology helps learning through what it enables teachers and learners to do. A quiz can reveal an error pattern quickly, but learning benefit depends on how that pattern changes the next teaching step. Evaluation therefore follows the full sequence from objective to evidence and response.

Fact 8.

$$\text{Learning need} \rightarrow \text{tool choice} \rightarrow \text{activity} \rightarrow \text{evaluation}$$

Why it holds. Plausible distractors can make a misconception visible in response patterns. A fresh question after targeted teaching tests improvement more directly than click counts.

Method 8.

1. Name the learner objective and the misconception or difficulty to investigate.
2. Choose a suitable accessible tool and design questions that reveal reasoning.
3. Inspect response patterns and adapt teaching accordingly.

4. Use a new question or response sample to judge whether understanding improved.

Recognising it. For an educational-technology proposal, state the learning purpose, diagnostic task, teacher response and evidence of value.

Pr 29. A short digital quiz can reveal misconceptions immediately, but the teacher must use the results to adapt instruction for it to support learning.

Solution. The brief digital quiz can make response patterns visible quickly. Plausible incorrect options help reveal which misconception may be present. The teacher must interpret those results and respond, for example with targeted explanation or discussion. A fresh question then checks learning; the number of clicks or time on screen alone does not establish educational value.

Common error. More screen time is not itself evidence of better learning.

In the examination. State the learning goal before the tool name and finish with evidence of learning; programming content is outside this course pack.

Pr 30. What is missing if a teacher runs a digital quiz but never uses its results?

Pr 31. Propose a simple technology use to diagnose misconceptions, and state how you would evaluate its value.

Pr 32. A digital quiz reveals a common misconception. Which action best turns the result into a learning benefit?

(a) Judge success only by how many students clicked

(b) Reteach the issue and check understanding with a new question

(c) Keep the response chart but make no teaching change

Hints

Use one hint at a time, then return to the problem.

Pr 3 (p. 3) (1) Select two distinct criteria, such as specialized preparation and ethical responsibility. (2) Explain each criterion briefly in your own words. (3) Attach one teaching example to each: show what the teacher learns or does and why that action illustrates the chosen criterion.

Pr 7 (p. 4) (1) Choose two roles beyond delivering subject explanations, ensuring they are meaningfully different. (2) For each role, name one observable action, such as diagnosing a learning gap or supporting strategy choice. (3) Explain how that action helps learning. A title without an action is only a partial answer.

Pr 11 (p. 5) (1) Set up two headings in your notes: before entry and early practice. (2) Under each, identify its main learning or support purpose. (3) Turn these into two parallel comparisons covering when support occurs and what it helps the teacher do. Do not classify solely by whether a workshop or mentor is present.

Pr 15 (p. 7) (1) Start with reviewing observation notes and student work rather than jumping directly to a new activity. (2) Identify one specific weakness and choose a change that addresses it. (3) Arrange the four-step cycle so the change is applied and then evaluated with new evidence. Keep every step linked to the same improvement need.

Pr 19 (p. 8) (1) Describe the need precisely: students give answers without sufficient explanation. (2) Set a focused goal and choose a relevant learning activity plus classroom application. (3) Specify a review period and compare response samples using clear explanation criteria. Ensure the evidence measures the original need.

Pr 23 (p. 10) (1) What aspects of teaching should your two items provide evidence for? Select two distinct aspects before choosing artifacts. (2) Consider a planning document and an anonymized learner-work sample; identify the different practice or outcome each can demonstrate. (3) Add a short explanation linking each artifact to its claim. Avoid claiming that a plan alone proves the intended learning occurred.

Pr 27 (p. 11) (1) Remove the personal or vague judgement from the original sentence. (2) Identify a specific observable difficulty and its effect on learners, using a clearly stated example rather than pretending to have evidence you did not collect. (3) Suggest one relevant next step in respectful language, such as clarifying success criteria and checking understanding before independent work.

Pr 31 (p. 12) (1) Choose a simple tool that can reveal reasoning or misconceptions, rather than starting with novelty. (2) Explain how the response pattern will guide a teaching action. (3) State how a new question or response sample will test improvement, while considering access and the defined learning objective.

Answers

Pr 2 (p. 3) No. Professional status also involves specialized preparation, knowledge, responsibility and organized standards of practice.

Pr 3 (p. 3) Specialized preparation: teachers study subject knowledge and pedagogy. Ethical duty: teachers assess fairly and protect learner dignity. Two marks for each criterion-example pair.

Pr 4 (p. 3) (a) Correct. Preparation and ethical responsibility are distinct criteria with clear teaching links.

Pr 6 (p. 4) It names an activity but does not explain how evidence is interpreted or used to support learning.

Pr 7 (p. 4) Assessor: identifies learning gaps using evidence and feedback. Learning guide: supports students in choosing strategies and solving problems. Two marks per role with a concrete action.

Pr 8 (p. 4) (b) Correct. The teacher supports the learner in selecting and evaluating a strategy.

Pr 10 (p. 5) No. A workshop during employment may be induction support or continuing development, depending on career stage and purpose.

Pr 11 (p. 5) Pre-service occurs before entry and develops foundational subject and pedagogical knowledge. Induction occurs during early practice and provides mentoring and support in the real workplace. One mark each for stage and purpose in both cases.

Pr 12 (p. 5) (c) Correct. Career stage and purpose define induction in this scenario.

Pr 14 (p. 7) It identifies an issue but does not show whether the teacher acted on it or improved practice.

Pr 15 (p. 7) Review evidence; identify one specific weakness; plan and apply a targeted change; evaluate the next lesson using evidence. One mark per coherent step.

Pr 16 (p. 7) (a) Correct. An action was taken, but its effect has not been assessed.

Pr 18 (p. 8) Participation. Improved teaching or learning needs additional evidence from practice and outcomes.

Pr 19 (p. 8) Need: weak explanations. Goal: improve evidence-based responses. Activity: learn and apply structured questioning. Evidence: compare student response samples before and after a defined period. One mark per linked component.

Pr 20 (p. 8) (b) Correct. This connects learning, classroom use and relevant evidence of impact.

Pr 22 (p. 9) No. Relevance, clear organization and reflection matter more than volume alone.

Pr 23 (p. 10) A lesson plan can demonstrate planning and alignment. An anonymized student-work sample with commentary can demonstrate learning and the effect of feedback. Two marks per item and explanation.

Pr 24 (p. 10) (c) Correct. A plan documents intentions and design; outcomes require additional evidence.

Pr 26 (p. 11) No. It can challenge practice, but uses evidence, respect and actionable suggestions.

Pr 27 (p. 11) Example: Several students could not identify the task goal; stating the success criteria and checking one example before independent work may help. Credit observable evidence, effect, actionable suggestion and respectful language.

Pr 28 (p. 11) (a) Correct. This reports an observable event that can inform an improvement discussion.

Pr 30 (p. 12) The evidence-to-action step: diagnosis has not been used to adapt instruction or check improvement.

Pr 31 (p. 12) Use a brief concept quiz with plausible distractors. Review response patterns, reteach the misconception and check with a new question. Evaluate whether explanations or transfer improve, not only whether students clicked. One mark each for tool, diagnosis, response and evaluation.

Pr 32 (p. 12) (b) Correct. This links diagnosis to action and evidence of improved understanding.

Professional Development Seminar, Test A

20 marks. Suggested time 20 minutes. Multiple-choice questions carry 1 mark each, written problems 4 marks each. Work without notes.

Part I. Choose one answer.

1. Which is a professional criterion emphasized in the lecture?

   (a) No responsibility to society
   (b) Learning stops at graduation
   (c) No preparation
   (d) Specialized knowledge and ethical standards

2. Using student work to improve instruction best illustrates the teacher as:

   (a) A storage clerk only
   (b) An assessor and learning guide
   (c) A person who never changes plans
   (d) A passive spectator

3. Support during a teacher first years of practice is mainly:

   (a) Only pre-service admission
   (b) Induction
   (c) Retirement
   (d) Waste management

4. The central value of supervised field experience is:

   (a) Eliminating evaluation
   (b) Replacing all theory
   (c) Connecting theory to practice with feedback
   (d) Avoiding learners

Part II. Show your reasoning.

1. Give two reasons teaching is considered a profession, each with an example.

2. Describe two teacher roles beyond explaining subject content.

3. Compare pre-service preparation with induction in two clear points.

4. Give a four-step improvement cycle for a trainee after an observed lesson.

Professional Development Seminar, Test B

20 marks. Suggested time 20 minutes. Multiple-choice questions carry 1 mark each, written problems 4 marks each. Work without notes.

Part I. Choose one answer.

1. Which is the strongest development goal?

   (a) Improve one identified practice and evaluate its effect
   (b) Collect the most certificates
   (c) Become better somehow
   (d) Attend anything

2. What makes a portfolio stronger?

   (a) No evaluation
   (b) More unrelated documents
   (c) Evidence linked to goals and reflection
   (d) Only decorative pages

3. Constructive professional feedback should be:

   (a) Only praise regardless of evidence
   (b) Unrelated to teaching
   (c) Personal and humiliating
   (d) Based on observation and useful next steps

4. The best reason to choose an educational tool is:

   (a) It supports a defined learning objective
   (b) It has the most animations
   (c) Everyone must use the same tool
   (d) It is newest

Part II. Show your reasoning.

1. Write a short development plan for a teacher whose students struggle to explain answers.

2. Name two portfolio items and explain what each can demonstrate.

3. Rewrite this feedback constructively: Your lesson was bad.

4. Propose a simple technology use to diagnose misconceptions, and state how you would evaluate its value.

Professional Development Seminar, Test A: answers

Suggested marking points are for self-study; your lecturer may use a different scheme.

Question 1. (d) Professional practice combines expertise, preparation and ethical obligations.

Written part. Specialized preparation: teachers study subject knowledge and pedagogy. Ethical duty: teachers assess fairly and protect learner dignity. Two marks for each criterion-example pair. (Section: p. 2.)

Question 2. (b) Assessment information guides the next teaching decision.

Written part. Assessor: identifies learning gaps using evidence and feedback. Learning guide: supports students in choosing strategies and solving problems. Two marks per role with a concrete action. (Section: p. 3.)

Question 3. (b) Induction bridges initial preparation and independent professional practice.

Written part. Pre-service occurs before entry and develops foundational subject and pedagogical knowledge. Induction occurs during early practice and provides mentoring and support in the real workplace. One mark each for stage and purpose in both cases. (Section: p. 4.)

Question 4. (c) Practice tests and develops professional knowledge in authentic settings.

Written part. Review evidence; identify one specific weakness; plan and apply a targeted change; evaluate the next lesson using evidence. One mark per coherent step. (Section: p. 6.)

Professional Development Seminar, Test B: answers

Suggested marking points are for self-study; your lecturer may use a different scheme.

Question 1. (a) A focused goal connects action to measurable change.

Written part. Need: weak explanations. Goal: improve evidence-based responses. Activity: learn and apply structured questioning. Evidence: compare student response samples before and after a defined period. One mark per linked component. (Section: p. 7.)

Question 2. (c) The reader can see what the evidence proves and how practice developed.

Written part. A lesson plan can demonstrate planning and alignment. An anonymized student-work sample with commentary can demonstrate learning and the effect of feedback. Two marks per item and explanation. (Section: p. 8.)

Question 3. (d) Evidence-based feedback supports improvement while respecting the colleague.

Written part. Example: Several students could not identify the task goal; stating the success criteria and checking one example before independent work may help. Credit observable evidence, effect, actionable suggestion and respectful language. (Section: p. 10.)

Question 4. (a) Pedagogical purpose should guide selection.

Written part. Use a brief concept quiz with plausible distractors. Review response patterns, reteach the misconception and check with a new question. Evaluate whether explanations or transfer improve, not only whether students clicked. One mark each for tool, diagnosis, response and evaluation. (Section: p. 11.)

Facts at a glance

1. Teaching as a profession p. 2

Criterion + explanation + teaching example

2. Professionalization and the teacher role p. 3

Role → observable professional action

3. Teacher formation: before and during service p. 4

Pre-service → induction → continuing development

4. Preparation, practice and professional standards p. 6

Knowledge → supervised application → feedback → improvement

5. Plan continuing professional development p. 7

Need → goal → activity → evidence of impact

6. Portfolios and reflective evaluation p. 8

Evidence + reflection + link to a goal

7. Ethics, collaboration and learning communities p. 10

Observation + effect + constructive next step

8. Educational technology and evaluation of impact p. 11

Learning need → tool choice → activity → evaluation

Notation

Symbols are listed by section. The same letter may name different quantities in different sections; read the section's Fact.

Sources

Elective lecture themes: teaching as a profession, teacher formation, professional learning, portfolios, standards and educational technology. This follows the teacher-development focus of the supplied file; it is not a generic career-skills course.

The source audit inventories 159 non-programming files; 95 yielded substantial extracted text. Scanned diagrams and handwritten or image-only material were not fully transcribed. Check lecturer-specific derivations and conventions. The practice tests are original study material, not predictions of an examination.

Index

Topics and quantities, with the page where each is introduced. Quantities show their symbol; the Index of notation lists symbols in alphabetical order.

E

Educational technology and evaluation of impact, 11

Ethics, collaboration and learning communities, 10

P

Plan continuing professional development, 7

Portfolios and reflective evaluation, 8

Preparation, practice and professional standards, 6

Professionalization and the teacher role, 3

T

Teacher formation: before and during service, 4

Teaching as a profession, 2

Index of notation

Latin letters first, then Greek letters, then differences (Δ). The same letter can name different quantities in different sections.